

Grade 6 Accelerated
Unit 13
Lesson 1 Practice

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1. What is the value of the expression $6 \cdot 2 - (x - 4)$ if $x = 1$?

2. Find the value of $8 - 2(4 + y) + 2y \cdot 2$ when $y = 3$.

3. Evaluate $2p + \frac{16}{j} - 11$ for $p = 5$ and $j = 8$.

4. Evaluate $10 - 20 \div 4f + 11$ if $f = 2$.

5. Find the value of $7 \cdot (2m - 4m + 18)$ when $m = 3$.

6. Find the value of $7 - (y + 12)^2 + 6m$ if $m = 6$ and $y = 2$.
7. Find the value of $12g - (16 - u - 8) \div 2$ if $g = 3$ and $u = 11$.
8. Evaluate $8a + 4b(b + 10)$ for $a = 3$ and $b = 2$.
9. What is the value of the expression $5g \cdot h^3 + (h - 1)^2$ if $g = 1$ and $h = 2$?
10. What is the value of the expression $\left(\frac{b^2-6}{4}\right) \cdot 7 + y^2$ if $b = 2$ and $y = 5$?

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Lesson 2 Practice

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1. Evaluate the expression $8b + 10 - (9 - 6)^2$ for $b = -5$.

2. Evaluate $-7m - 5q^2 + \frac{s}{2}$ when $m = -8$, $q = -2$, and $s = 56$.

3. Evaluate $2s - 6q - f^2$ for $s = -8$, $q = 10$, and $f = -12$.

4. Evaluate $-m^3 - 3d + (n - y)$ if $m = 4$, $d = -6$, $n = -9$, and $y = 0$.

5. Evaluate $4g^4 - \frac{z}{4}$ when $g = -4$ and $z = 24$.

6. What is the value of $5(8x + 4) - 3m^2$ if $x = 3$ and $m = 15$?
7. Find the value of $-9(h - 6) + 2a - (9 - a)^5$ if $h = 6$, and $a = 12$.
8. Evaluate the expression $p^3 - 5k(y - k)$ for $p = 6, k = -4$ and $y = 1$.
9. Evaluate the expression $8 \div j - 12 \div (-10 + 4)$ if $j = -4$.
10. What is the value of the expression $4m - 10 + (a - 11) \cdot 4^3$ if $m = 5$ and $a = -1$?

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Lesson 3 Practice

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Determine if each pair of expressions are equivalent.

1. $-2(10 + 4)$ and $-2(10) + 2(4)$

2. $0.5(5 - 2)$ and $0.5 \cdot 5 + 0.5 \cdot -2$

3. $\frac{3}{4} \times \frac{9}{10}$ and $\frac{9}{10} \times \frac{3}{4}$

4. $\left(\frac{2}{3} + \frac{1}{8}\right) + \frac{1}{6}$ and $\frac{2}{3} + \left(\frac{1}{8} + \frac{1}{6}\right)$

5. $\frac{1}{6} + \frac{2}{9}$ and $\frac{2}{9} + \frac{1}{6}$

6. $\frac{7}{11} - \frac{1}{3}$ and $\frac{1}{3} - \frac{7}{11}$

7. $\frac{5}{7} + \left(\frac{1}{2} + \frac{3}{4}\right)$ and $\left(\frac{5}{7} + \frac{1}{2}\right) + \frac{3}{4}$

8. $\left(\frac{1}{5} - \frac{2}{5}\right) - \frac{8}{9}$ and $\frac{1}{5} - \left(\frac{2}{5} - \frac{8}{9}\right)$

9. Rewrite $(6x - 2y) + 9$ using the associative property.

10. Rewrite $9z - 7b$ using the commutative property.

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Lesson 4 Practice

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1. Rewrite the expression $(-5h + 8) + 2h$ using the associative property.

2. Rewrite the expression $(-5h + 8) + 2h$ using the commutative property.

3. Rewrite the expression $8 \cdot (-7 \cdot 3y)$ using the associative property.

4. Rewrite the expression $8 \cdot (-7 \cdot 3y)$ using the commutative property.

5. Rewrite the expression $-2r + 4(11k + 3)$ using the distributive property.

6. Rewrite the expression $-2r + 4(11k + 3)$ using the commutative property of multiplication.
7. Rewrite the expression $-2r + 4(11k + 3)$ using the commutative property of addition.

Write two equivalent algebraic expressions for each expression.

8. $(2 + 4) \cdot 7$ 9. $7t \cdot 2$ 10. $-6x + 6 - 18$

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Lesson 5 Practice

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Write an algebraic expression for each written description.

1. The quotient of a number and nine.

2. The product of a number and eleven.

3. Three-fifths more than a number.

4. A number doubled.

5. The difference of a twenty and a number.

6. The school provides each student, s , 5 hall passes per quarter.
7. There are 8 snack bags, each with b , number of snacks in each bag.
8. A hair salon makes \$18 per haircut, h .
9. An electricity company charges \$0.11 per kilowatt, k .
10. A football team scored 14 points less than the other team, t .

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Write an algebraic expression to represent each real-world situation.

1. Julia has 11 video games downloaded to her device. She deleted x of those games from the device. How many video games are left on her device?

2. Tony has x tickets from a local arcade. His brother has 10 times more tickets than Tony. How many tickets does his brother have?

3. Susan sold 42 candy bars for a fundraiser. Her friend sold x more candy bars than her. How many candy bars did the friend sell?

4. A grocery store has received a shipment of 60 new boxes of cereal to be placed equally on x shelves. How many boxes will be on each shelf?

5. Harold has a plant that measured h inches tall last week. This week, the plant is 19 inches tall.

For each expression, create a real-world scenario that the algebraic expression could represent.

6. $x + 7$

7. $\frac{22}{x}$

8. $71y$

9. $k - 81$

10. $12 + h$

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For questions 1-2, draw a representation of the expression with algebra tiles. Cross out any zero pairs. Then write an equivalent algebraic expression after eliminating any zero pairs.

	Algebraic Expression and Representation	Equivalent Algebraic Expression
1.	$-8x - 2 + 6x$ Representation:	
2.	$10 + z - 6 + z$ Representation:	

Write an equivalent expression by combining like terms for questions 3- 10.

3. $4x + 9x$

4. $-2y + 5 + y - 7$

5. $r + -r + r - r$

6. $6y - 3 + 8 - 4y$

7. $y - 4 + 5y + 8$

8. $10x + 9 + x - 1$

9. $-5y + 2 + 3 + 9y$

10. $8 + z - 10 + 2z$

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Lesson 8 Practice

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Find the sum of each expression using the fewest terms possible.

1. $(0.5x + 0.1) + (0.6x - 0.9)$

2. $\left(-\frac{2}{5} + 7y\right) + \left(4y - \frac{3}{5}\right)$

3. $(-0.3x) + (x + 9)$

4. $\left(5\frac{1}{2}y - \frac{4}{5}\right) + \left(8\frac{1}{5} - 3\frac{3}{4}y\right)$

5. $\left(10 - \frac{2}{3}y\right) + \left(-7 + \frac{2}{3}y\right)$

Find the sum of each expression using the fewest terms possible.

6. $(2.7 + 8.4x) + (-3.6x + 7.5) =$

7. $\left(\frac{1}{8}x + \frac{3}{4}\right) + \left(\frac{3}{4}x - \frac{7}{16}\right) =$

8. $(-2.9y + 6) + (5.9 + 4y) =$

9. $\left(1.0 + \frac{4}{9}x\right) + (-6.6 + x) =$

10. $\left(11.8 + \frac{3}{7}x\right) + \left(-\frac{11}{7} + x\right) =$

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Lesson 9 Practice

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Find the difference of each expression using the fewest terms possible.

1. $(0.6x + 0.7) - (0.8x - 0.9)$

2. $\left(-\frac{4}{5} + 9y\right) - \left(3y - \frac{3}{5}\right)$

3. $(-0.2z) - (z + 8)$

4. $\left(8\frac{1}{2}y - \frac{2}{5}\right) - \left(6\frac{3}{5} - 2\frac{3}{4}y\right)$

5. $\left(10 - \frac{4}{9}q\right) - \left(-8 - \frac{2}{9}q\right)$

Simplify each of the expressions.

6. $(7.6 - 4.9x) - (2.8x - 3.7) - (9.8x - 4.2)$

7. $\left(\frac{1}{8}x - \frac{3}{4}\right) + \left(\frac{3}{4}x + \frac{13}{16}\right) + \left(\frac{3}{8}x - \frac{1}{4}\right)$

8. $(2.8y + 8) - (2.7 + 3y) + (-8.7 + 5y)$

9. $(2.8 + \frac{4}{9}r) + (-6.8 + r) - \left(-4.7 + \frac{7}{9}r\right)$

10. $\left(\frac{5}{6} + \frac{3}{7}n\right) - \left(-\frac{1}{6} + n\right) - \left(2\frac{5}{6} + 7\frac{2}{7}n\right)$

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Lesson 10 Practice

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Determine if the expressions are equivalent for the given values.

	Value	$8x + 4x - 5$	$2(6x - 8) + 11$	Equivalent?
1.	$x = -4$			☐ yes ☐ no
2.	$x = 0$			☐ yes ☐ no
3.	$x = 0.5$			☐ yes ☐ no
4.	$x = \frac{1}{3}$			☐ yes ☐ no

5. Depending on the value being substituted into the expressions, the evaluated expressions
☐ always
☐ sometimes
☐ never
 had the same value as each other.

Determine whether the expressions are equivalent for the given values.

		$\frac{1}{5}(15 + 20y) + 2x$	$4\left(2x + y + \frac{3}{4}\right)$	Equivalent?
6.	$x = -2$ $y = 6$			☐ yes ☐ no
7.	$x = 0$ $y = 1$			☐ yes ☐ no
8.	$x = \underline{\hspace{1cm}}$ $y = \underline{\hspace{1cm}}$			☐ yes ☐ no

Determine if the following algebraic expressions are equivalent using any method you choose.

	Expression A	Expression B	Equivalent?
9.	$x(13 - 15) - \frac{3}{2}(4 + 6x)$	$-11x - 15$	☐ yes ☐ no
10.	$\frac{5}{7}(2x + 12x) - (6^2)$	$-36 + 10x$	☐ yes ☐ no

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From the specified set of integers, determine which, if any, make the equation true.

$\{-10, -4, -1, 0, 1, 3, 12\}$

1. $20 - 5w = 10w + 5$

3. $3y + 5 = 4y + 15$

2. $4 + x = 2x - 8$

4. $2p - 6 = 6p - 18$

Determine if the inequality $18 < 2m - 6$ is true for each value.

5. 8

6. 12

7. 15

Determine if the inequality $-y + 3 + 8y > -3y - 7$ is true for each value.

8. -5

9. 0

10. 5

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Write an equivalent expression using the associative, commutative, or distributive property. State the property used.

1. $6x - \frac{2}{7}(21x - 35)$

2. $(11x + 15) - 3x + 7$

3. $\left(-\frac{3}{5}y + 2\right) \cdot 20$

4. $-24\left(\frac{3}{4}x + 1\right) + (18 - 3x)$

5. If expressions are not equivalent, then they will

- ☐ always
- ☐ sometimes
- ☐ never

have the same terms when they are simplified using properties of operations.

Use properties of operations to determine if the expressions are equivalent.

6. $0.5x - 0.1(5x - 2)$ and $7x - (-2 + 7x)$

7. $(4x + 12) - 6$ and $7x - 3(x - 2)$

8. $2\frac{1}{5}x - 8 + x$ and $\frac{16}{5}\left(x - \frac{5}{2}\right)$

9. $0.3 + 0.1(17x - 4)$ and $2x - \left(\frac{3}{5}x - \frac{2}{5}\right)$

10. Expressions that are equal are not equivalent.

- ☐ True
- ☐ False